



AKSHITA DHIMAN
Bachelor of Technology
in Computer Science And Engineering
Indian Institute of Technology Ropar

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LinkedIn

EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	7.55 (till 3rd Sem)	2024-2028
Senior Secondary	Central Board of Secondary Education	89%	2024
Secondary	Central Board of Secondary Education	96%	2022

PROJECTS

- **Instruction Level Parallelism Analyzer** Jan 2026 - May 2026
Computer Architecture **Github**
 - Built a performance analysis tool that processes real execution traces collected via **Intel Pin (PinTool)**
 - Designed backend logic to analyze instruction streams and compute **ILP metrics**
 - Enabled users to upload and analyze program traces through a **browser-based interface**
- **C Code Optimizer** Jan 2026 - May 2026
Programming Paradigms **Github**
 - Built a **code optimization tool** with a web interface for analyzing and transforming C programs
 - Developed visualization features to display **Control Flow Graph (CFG)** for debugging and analysis
 - Implemented **dead code elimination**, **constant folding**, and CFG-based optimizations
- **LSM Trees** Aug 2025 - Dec 2025
Data Structures & Algorithms **Github**
 - Developed a **log-structured storage engine** with a web-based interface for key-value data interaction
 - Implemented core components including **memtables**, **SSTables**, and **compaction**
 - Enabled visualization of storage structure and data flow

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Verilog
- **Backend & Core Development:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Functional Programming, API Design, Data Parsing & Processing, Performance Optimization
- **Systems & Computer Science:** Operating Systems, Computer Architecture, Instruction-Level Parallelism, Compiler Basics, Memory Management
- **Tools & Technologies:** Linux (Ubuntu), Git, Intel Pin (PinTool), VS Code, Debugging Tools
- **Web & Visualization:** HTML, CSS, JavaScript, Data Visualization (Graphs, Timelines)

RELEVANT COURSEWORK

- **Computer Science:** Data Structures & Algorithms, Discrete Mathematics, Computer Architecture, Digital Logic Design, Programming Paradigms

MISCELLANEOUS

- **Inter IIT Cultural Meet 7.0**, Secured 3rd place in Stage Play competition
- **Athletics**, Won 1st place in 400m relay in IYSC
- **Music**, Participated in multiple band competitions